Executive Summary

Predicting Diabetes Readmission Within 30 Days

Github: https://github.com/Erdos-Projects/fall-2026-hospital-readmission-for-diabetes

Data: https://archive.ics.uci.edu/dataset/296/diabetes+130-us+hospitals+for+years+1999-2008

Background and Project Overview. Diabetes requires ongoing management of blood glucose, medications, and related conditions. For patients hospitalized with diabetes, readmission soon after discharge can interrupt recovery and add to healthcare costs. Predicting who will return is difficult because 30-day readmissions are relatively uncommon, and information recorded during a hospital stay cannot capture every factor that shapes recovery after discharge. Although not every readmission is preventable, a useful risk model could help care teams target follow-up support. This project examines whether encounter information can identify patients at greater risk of readmission within 30 days.

Stakeholders. Patients hospitalized with diabetes and their families are the central stakeholders. Clinicians, discharge planners, and care managers could use estimates to prioritize follow-up after discharge.

Dataset, Data Cleaning, and Data Preprocessing. The dataset contains 101,766 hospital encounters and 47 recorded features. Because a patient can have multiple encounters, we retained the patient identifier to keep that patient's records in the same data split, then excluded it from the model inputs. During cleaning, we removed the encounter identifier, treated "?" and "NULL" as missing values, standardized column names, and translated coded admission and discharge fields into descriptive categories. Each model's preprocessing pipeline handled missing predictor values. We classified readmission within 30 days as the positive outcome. Readmission after 30 days and no recorded readmission formed the negative outcome.

Main Question. Can one develop an interpretable model using original and engineered features that match or exceed the F2 score of the baseline model?

KPIs and Metrics. In this dataset, 11.16% of encounters were followed by a recorded readmission within 30 days. In a separate 2011 study[1], readmissions following hospitalizations for diabetes with complications among Medicaid patients ages 18–64 accounted for an estimated $251 million in hospital costs across 23,700 readmissions, or about $10,600 per readmission. Because we prioritize identifying patients who may be readmitted, we use F2 as our primary metric and report precision to show the impact of false alerts.

Modeling Process and Methodology. We first divided the dataset into a development set of 81,412 encounters (80%) and a test set of 20,354 encounters (20%). We then divided the development set into 65,129 training encounters and 16,283 validation encounters. For both divisions, we used a single split produced by sklearn's StratifiedGroupKFold. We grouped encounters by patient identifier so that encounters from the same patient do not appear on both sides of a split, while approximately preserving class balance of 30-day readmissions in each

[1] Hines, Anika L., et al. "Conditions with the Largest Number of Adult Hospital Readmissions by Payer, 2011." HCUP Statistical Brief, no. 172, Agency for Healthcare Research and Quality, Apr. 2014, hcup-us.ahrq.gov/reports/statbriefs/sb172-Conditions-Readmissions-Payer.jsp.

subset. We fitted candidate models on the training subset and used the validation subset to compare model configurations and select the classification threshold that maximized F2 for each model. We considered models such as LightGBM, random forest classifier, XGBoost classifier, CatBoost, and logistic regression. The final test fits the chosen candidates on the development set and evaluates it on the test set using its a validation-selected threshold.

Baseline Model. The baseline is a logistic regression using all 47 available predictors. It performs minimal preprocessing. We fitted the pipeline on the training subset and selected the threshold that maximized F2 on the validation subset. We then refitted the entire pipeline on the 81,412-encounter development set and evaluated it on the test set using that fixed threshold.

Light Logistic Regression Model. The light model uses seven original features: prior inpatient visits, prior emergency visits, payer code, discharge disposition, primary diagnosis, number of diagnoses, and number of medications. It groups primary diagnosis codes, imputes numerical values by the median, scales them, fills missing categorical values with a Missing category, one-hot encodes categories, and fits class-weighted logistic regression

Heavy Logistic Regression Model. The heavy model uses 124 original and engineered features. These include encounter characteristics, medication information, grouped diagnoses, transformed numerical measures, and interactions between variables. We selected the inputs and model settings through an Optuna search that maximized validation F2. We included this model to test whether extensive feature engineering improved on the seven-input model.

Results. At thresholds selected on the validation set, the heavy logistic regression model ranked first on validation, but F2 score of 0.43951 on its best threshold was not convincing enough to suggest it was the better model. Therefore, we chose the light logistic regression as our final model. It achieved a test F2 score 0.42477 beating the baseline which got an F2 score of 0.41277. This gives a positive answer to our main question on this test set: a small set of recorded variables some preprocessing matched and modestly exceeded the baseline F2 score. The light model's short input list is easier to inspect. Further analysis on its weights is warranted. At its chosen threshold, the light model had 14.7% precision and 80.8% recall. Across 20,354 test encounters, it produced 1,917 fewer false positives than the baseline while identifying 116 fewer actual readmissions.

Limitations and Future Work. The final model's modest improvement in F2 suggests that a small set of features can capture useful information about readmission risk. Before considering clinical use, we would evaluate the model on more recent data from other hospitals, check whether its predicted probabilities are well calibrated, and examine performance across patient groups. Developing medication-specific models is another possibility, but each group needs enough readmissions to support reliable training and evaluation. For example, encounters involving first-generation sulfonylureas accounted for only nine readmissions. Future work could test larger treatment groups and determine if separate models improve performance. Another possibility is to develop models for each age group.